

ORF 245: Fundamentals of Statistics – Spring 2025

Precept 5

1. Random Sample

- Independent and Identically Distributed R.V.'s.: $X_1, \dots, X_n$ are independent and have the same distribution (population distribution). Let $\mu = \mathbb{E}[X_1]$ and $\sigma^2 = \mathbb{V}[X_1]$.
- Sample mean $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$. $\mathbb{E}[\bar{X}] = \mu$, $\mathbb{V}[\bar{X}] = \sigma^2/n$.
- Law of Large Number: $\bar{X} \rightarrow \mu$
 - Bias-and-variance decomposition $\mathbb{E}[|\bar{X} - \mu|^2] = |\mathbb{E}[\bar{X}] - \mu|^2 + \mathbb{V}[\bar{X}] = \sigma^2/n$.
 - Statistical implication: use the sample mean to estimate the population mean.
- Central limit theorem:
 - $\frac{\sqrt{n}(\bar{X} - \mu)}{\sigma} \xrightarrow{d} \mathcal{N}(0, 1)$.
 - Unknown σ^2 : $\frac{\sqrt{n}(\bar{X} - \mu)}{\sqrt{S^2}} \xrightarrow{d} \mathcal{N}(0, 1)$, S^2 is the sample variance.
 - Statistical implication: uncertainty quantification (hypothesis test, confidence interval).

2. An important assumption: missing at random.

- Suppose you are interested in estimating some quantity μ for the population (degree of satisfaction, support rate) and consider the following sampling methods.
 - (a) $\bar{X}^{(1)}$: You send questionnaires to n_1 people forcing them to respond to the questionnaire anonymously and then do the average over n_1 response.
 - (b) $\bar{X}^{(2)}$: You send questionnaires to N people, only n_2 response and you do average over the n_2 response.

Will the two methods provide a good estimate of μ ?

- **Solution:** For the first method, we have $\bar{X}^{(1)} \xrightarrow{p} \mu$.

For the second method: Let R be 1 if one will respond and 0 if one doesn't respond. We then have

$$\bar{X}^{(2)} = \frac{\frac{1}{N} \sum_{i=1}^N X_i R_i}{\frac{1}{N} \sum_{i=1}^N R_i} \xrightarrow{p} \frac{\mathbb{E}[XR]}{\mathbb{E}[R]}.$$

- (a) Is a good estimate if X and R are uncorrelated (missing at random). It will have a problem if the two quantities are correlated: it is “biased”.
- (b) Random sampling is the best (in terms of bias, MSE), validated theoretically and empirically. But it may be hard to achieve in practice.
- (c) Remedies: R and X are approximately independent given observed variable Z (causal inference, untestable assumption, require domain knowledge).

3. Central limit theorem (CLT) Uncertainty quantification using CLT.

- (a) Example 1 (Calculating the approximate probability)

A high stakes national university entrance exam in India is taken by more than 95% of high school seniors (a lot of people!). For a randomly selected senior, the score is expected to be 500 with a standard deviation of 80. Approximately, what is the probability that the average score of 100 randomly sampled students will be 490 to 510? What's the probability of scoring above 510?

• **Solution:**

Let X_i be the score of student i .

By CLT, $\bar{X} \stackrel{a}{\sim} N(500, 80/\sqrt{100})$.

Standardizing, this yields:

$$\mathbb{P}\left[\frac{1}{100} \sum_{i=1}^{100} X_i \in (490, 510)\right] = \mathbb{P}[-1.25 < Z < 1.25] = 1 - 2\Phi(-1.25) = 0.79.$$

The probability of scoring above 510 is $\Phi(-1.25) \approx 0.105$.

(b) Example 2 (Calculating the minimum sample size to achieve certain confidence)

Princeton's basketball team expects to win 60% of its games. Assuming ties do not occur, approximately how many games should they play before they can be on average 95% confident of winning more than half of them?

• **Solution:**

$X_i \sim \text{Ber}(0.6)$, so $\mu = 0.6$ and $\sigma^2 = 0.24$. We want to find n such that

$$\mathbb{P}\left(\bar{X} > \frac{1}{2}\right) \geq 0.95$$

By the CLT, therefore

$$\mathbb{P}\left(\frac{\bar{X} - 0.6}{\sqrt{0.24/n}} > \frac{0.5 - 0.6}{\sqrt{0.24/n}}\right) \geq 0.95$$

$$\mathbb{P}\left(\mathcal{N}(0, 1) > \frac{0.5 - 0.6}{\sqrt{0.24/n}}\right) \gtrsim 0.95$$

$$\mathbb{P}\left(\mathcal{N}(0, 1) < \frac{0.1}{\sqrt{0.24}}\sqrt{n}\right) \gtrsim 0.95$$

$$\frac{0.1}{\sqrt{0.24}}\sqrt{n} \gtrsim \Phi^{-1}(0.95)$$

$$\sqrt{n} \gtrsim 10\sqrt{0.24} \cdot \Phi^{-1}(0.95)$$

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$$\sqrt{n} \gtrsim 8.059$$

$$n \gtrsim 65$$